

Quadrilateral Support Kernels and Certified Normal-Disc Discovery

Exact cone reduction, support-dependent height bounds,
rank-sensitive search, and a barrier to uniform sparsity

Research continuation for Vladimir Reshetnikov's ProveIt project
Prepared with OpenAI ChatGPT; proofs, code, and independent audits included

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Abstract

The maintained unknot-recognition project now has compressed, independently checked operations on a supplied normal surface, but finding a suitable surface remains a separate task. We implement an exact discovery layer for a specified compatible quadrilateral sector and derive explicit support-sensitive bounds. If at most one quadrilateral type is allowed in each of k tetrahedra, contracting the triangle matching equations with identically zero quadrilateral contribution reduces the standard normal cone, modulo independent vertex-link factors, to at most $9k$ variables and $4k$ equations. The triangle block is a directed incidence matrix and every quadrilateral column has ℓ_1 norm at most four. A determinant expansion therefore bounds every primitive standard vertex triangle coordinate by 4^k , and every positive quadrilateral coordinate by 4^{k-1} , independently of the total number t of tetrahedra.

Two exact enumeration methods operate on this reduction: positive-support enumeration in $2^{9k} \text{poly}(t, k)$ time and a potential-arrangement method whose number of linear systems depends on the quadrilateral matching nullity d . Adapting the classical canonical Euler-characteristic convexity argument, a preliminary Q-ray search tests only $\binom{k}{d-1}$ systems for $d > 0$ to decide whether the sector contains a canonical positive-Euler surface. Every positive disc claim is checked by the existing weighted orbit machinery. Exhaustion has explicitly restricted meanings.

The implementation agrees with independent Regina enumerations on 1,718 sectors in each phase, with 962 complete discovery-certificate replays and 60 rejected mutations. We also derive a limitation: the Fibonacci layered solid tori force $\Omega(t)$ occupied quadrilateral tetrahedra for every essential standard vertex disc. Sparse search is consequently a conditional quasi-polynomial route, not a general solution. Dense sectors of nullity one remain tractable and are included in the experiments. The package provides proofs, portable Python code, exact fixtures, raw measurements, and integration instructions. It does not establish a general quasi-polynomial unknot recognizer.

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1 Outcome, source baseline, and relation to earlier work

1.1 The concrete advance

This continuation moves from evaluating a supplied normal vector to actually constructing candidate vectors. The input is a finite triangulation $\mathcal{T}$ of a compact orientable three-manifold with one torus boundary, together with allowed quadrilateral types. The output is either an independently checked essential-disc witness or a precisely delimited statement about that sector. An optional outer search enumerates small allowed supports.

The main mathematical refinement is the dependence on the number of allowed quadrilateral types, rather than all standard coordinates. A triangulation may contain many tetrahedra through which only triangles propagate. These tetrahedra still have to be read and their geometric data retained, but they need not contribute independent variables to the difficult cone search.

The three principal proved results are as follows.

1. An exact algebraic quotient of a compatible standard cone has at most $9k$ variables and $4k$ matching equations; an explicit integer-linear expansion reconstructs the original coordinates (Theorem 3.2).
2. Primitive standard vertex coordinates satisfy the sharper bounds $a_c \leq 4^k$ and $q_i \leq 4^{k-1}$, with a data-dependent product bound available (Theorem 4.3).
3. On Fibonacci layered solid tori, every essential standard vertex disc needs linear quadrilateral support (Theorem 9.1). This shows why the support bound alone cannot settle the general complexity objective.

The exact search and certificate code make these statements executable. The distinction between an algebraic candidate and an essential disc is preserved throughout.

1.2 Pinned repository state

The source snapshot was read through the selected GitHub connection at repository commit

```
fdeb5b1a20b84b93cb150e0232031837bb7d30cb.
```

The latest relevant unknot commit in that snapshot is

```
3a6c7e99fe0ea9fc37ab505d261e0d747acb8cf2.
```

The repository location is <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Topology/UnknotRecognition>. The snapshot already includes sparse incidence, independently certified weighted normal components, the quadrilateral disc-count core, and ordered compressed group-certificate replay. The maintained synthesis describes these in `synthesis/sparse_incidence.tex`, `synthesis/weighted_components.tex`, and the recent elimination sections. The retrieved root README contains older historical test counts; the recent modules and commit records are the appropriate baseline for this continuation.

We reuse the existing finite-manifold validator, integer coordinate encoding, normal arc geometry, weighted orbit producer, and independent disc-count checker. Two new modules are additive:

```
fastunknot/normal_sector.py
fastunknot/normal_sector_verify.py.
```

No claim in this report depends on having rerun the entire historical 1,090-test suite. The reported validation is the new, specifically described oracle and certificate audit. All bundled upstream runtime files are pinned and unchanged; the integration patch adds only the new modules and tests.

1.3 Attribution and the scope of originality

Normal matching equations, the ambiguity of triangle coordinates up to vertex links, canonical lifting, and the relationship between Q-vertices and standard vertices are established theory. Tollefson’s Q-theory supplies the foundational coordinate reduction [1]. Burton gives canonical extension and conversion between quadrilateral and standard solution sets [2]; in particular, his Corollary 4.4 and Lemma 4.5 give the canonical nature of non-link standard vertices and the standard extremality of canonical Q-vertices. Burton’s quadrilateral–octagon paper explicitly uses potentials on the vertex-link graph and proves the convexity of canonical Euler characteristic [3, proofs of Theorems 4.5 and 5.5]. Burton–Ozlen develop exact type search and rank-sensitive determinant arithmetic [4, 5].

We therefore present the graph contraction, explicit $9k/4k$ bounds, incidence minor refinement, and their executable combination as a support-sensitive derivation and implementation for this project. We do not claim priority over all literature for these numerical refinements. The standard ingredients and the new derivations are identified individually rather than grouped under a claim of a new general normal-surface theory.

1.4 The quasi-polynomial target in the current literature

Lackenby’s 2021 lecture announces quasi-polynomial unknot recognition and describes compressed hierarchies and accelerated hierarchy operations [11]. His July 2026 preprint proves a new hierarchy algorithm and develops useful certificate machinery [10]. Section 9 explicitly leaves some steps without sufficiently detailed running-time estimates. Theorem 14.4 provides efficient compressed operations for a supplied normal surface; that is different from an end-to-end bound on finding the required hierarchy.

The present work has the same important separation. An efficient fixed-sector discovery operation is one algorithmic component. A quasi-polynomial bound for all knot diagrams additionally needs control of sector selection, changes of triangulation, peripheral provenance, and subsequent geometric operations.

2 Coordinates, sectors, and the exact search contract

2.1 Finite triangulations and normal coordinates

Let $\mathcal{T}$ have t tetrahedra. Face identifications are affine maps encoded by permutations of the four local vertices. We allow loops, multiple identifications, repeated global vertices within one tetrahedron, and gluings between two distinct faces of the same tetrahedron. Every unpaired face lies in the boundary. The existing validator checks connectedness, orientability, edge links, finite vertex links, and that the single boundary component is a torus. Ideal vertices are outside this implementation’s domain.

Write $a_{i,v}$ for the triangle coordinate at local vertex v of tetrahedron i , and $q_{i,j}$ for its three possible quadrilateral coordinates. An integral vector is admissible when all coordinates are nonnegative, the face matching equations hold, and at most one quadrilateral type is nonzero per tetrahedron. The normal realization theorem identifies such vectors with embedded normal surfaces up to normal isotopy [6, Theorem 5.1].

Each face matching equation compares the number of arcs cutting off one face corner. In local notation it is

$$a_c + q_{\alpha(e)} = a_d + q_{\beta(e)}, \quad a_d - a_c = b_e(q) := q_{\alpha(e)} - q_{\beta(e)}. \quad (1)$$

An absent or forbidden quadrilateral contributes zero. When the two displayed quadrilaterals are the same variable, they cancel. This convention is essential for self-gluings.

Definition 2.1 (Allowed sector). A sector S selects one allowed quadrilateral type in each of k distinct tetrahedra and forces every other quadrilateral coordinate to zero. The k selected coordinates remain nonnegative and are allowed to vanish. Denote the resulting real standard-coordinate cone by C_S .

Thus k is an allowed-support size, whereas a particular candidate may have smaller actual support. Since forbidden coordinates are fixed to zero, C_S is a face of the nonnegative matching cone. Its extreme rays are therefore standard extreme rays of that cone. Every vector in C_S satisfies the quadrilateral compatibility condition.

2.2 Vertex links and primitive rays

For a global vertex v , let L_v put one triangle at every local corner representing v and zero elsewhere. Its normal surface is a sphere for an interior vertex and a disc for a boundary vertex. Therefore

$$\chi(L_v) = 2 \quad \text{or} \quad 1. \quad (2)$$

A primitive ray vector is a nonzero integral vector whose coordinate gcd is one and which spans an extreme ray. Triangle-only rays are precisely vertex links. They cannot be essential discs.

A primitive extreme-ray surface is connected. Indeed, if it were a disjoint union, its component vectors would be nonzero integral vectors in the same cone summing to the ray vector. Extremality makes every component a positive scalar multiple of it. Primitivity forces an integral scalar: an integer linear combination of its coordinates equals one, so any scalar producing integral coordinates is itself integral. Two positive such scalars cannot sum to one. This argument includes one-sided surfaces.

2.3 Statements the interface is allowed to make

Table 1 fixes the semantics before any performance discussion. An essential-disc witness concerns the supplied manifold; it becomes an unknot certificate only when that manifold is independently bound to the input knot exterior. The manifold validator checks its topology locally, but it does not prove that correspondence.

Status	Exact meaning
DISC_FOUND	The supplied triangulation contains a verified essential disc component in the returned normal vector.
NO_POSITIVE_EULER	The sector contains no canonical normal surface with positive Euler characteristic. In particular it contains no essential normal disc.
POSITIVE_EULER_ONLY	The complete Q-ray phase found positive Euler characteristic, but none of its tested rays was an essential disc.
NO_VERTEX_DISC_IN_SECTOR	Complete standard-ray enumeration found no essential standard vertex disc in this sector.
NO_VERTEX_DISC_UP_TO_SUPPORT	The complete outer search found no essential standard vertex disc using at most the requested support size.
INCONCLUSIVE	A configured allowance was exhausted; no exhaustion certificate or negative conclusion is issued.

Table 1: Search results are scoped to the supplied sector or triangulation. None of these strings claims a correspondence with a knot diagram.

3 An exact algebraic support kernel

3.1 The triangle difference graph

Construct a multigraph with one vertex for each of the $4t$ triangle coordinates. Every face-corner matching equation gives an oriented edge from c to d , labelled by the integer linear form b_e in (1). Reversing the orientation negates the form. The graph's connected components are exactly the global vertex classes, since both equivalence relations are generated by the same face-corner identifications.

Restrict all labels to the allowed variables in S . Contract every edge whose label is identically zero. The equation on such an edge requires equality of its endpoint triangle coordinates, so this operation is exact over both $\mathbb{R}$ and $\mathbb{Z}$. Keep every nonzero-label edge after contraction. In particular, a resulting loop must be retained: its triangle part vanishes, but it may impose a nontrivial equation on the quadrilaterals.

Lemma 3.1 (Only active quadrilateral incidences survive). *At most $4k$ matching equations have a nonzero restricted quadrilateral label. Every retained quadrilateral column has ℓ_1 norm at most four.*

Proof. A normal quadrilateral meets each of its tetrahedron's four faces in one normal arc. Consequently its coordinate occurs in at most four face-corner matching equations, once on each glued face; a boundary face gives no matching equation. Every nonzero restricted row has at least one allowed occurrence. There are at most $4k$ occurrences in total. If both sides of a face equation refer to the same allowed variable, their signs are opposite and they cancel. No coefficient of magnitude two is created. Row deletion and cancellation can only decrease each column's absolute sum. $\square$

Theorem 3.2 (Support kernel). *For a sector of size k , the cone C_S admits an explicit isomorphism*

$$C_S \cong K_S \times \mathbb{R}_{\geq 0}^u, \quad K_S = \{(h, q) \geq 0 : M_S(h, q) = 0\}, \quad (3)$$

where the last factors are independent vertex-link coordinates, K_S has at most $9k$ variables, and M_S has at most $4k$ rows. Every row has Euclidean norm at most two. The isomorphism and its inverse preserve integer points, primitive rays, and extreme rays. Expansion to the original $7t$ coordinates copies retained triangle-class values and inserts zeros or vertex-link values.

Proof. Let p_v be the number of triangle classes remaining at a global vertex v . The contracted graph on them is connected. If $p_v = 1$, its one triangle variable cancels from every incident equation, including any loops; it is an independent nonnegative vertex-link factor. Remove that variable, retaining all loop equations on q .

For $p_v \geq 2$, let m_v be the number of retained edges in that component. Connectivity gives $p_v - 1 \leq m_v$, and hence $p_v \leq 2m_v$. If p is the total number of retained triangle-class variables and m is the number of retained equations, then

$$p = \sum_{p_v \geq 2} p_v \leq 2 \sum_v m_v = 2m \leq 8k.$$

Adding the k quadrilateral variables gives at most $9k$ variables. A retained row is one triangle difference minus at most two allowed quadrilaterals. Its nonzero entries are at most four copies of 1 or -1 , so its Euclidean norm is at most two.

All triangle coordinates in a contracted class must be equal. Conversely, copying a class value to its original corners satisfies every contracted zero-label equation, and the retained rows are exactly the remaining matching equations. A removed singleton class can take any nonnegative value,

independently of q ; this is precisely a vertex link. These observations give mutually inverse integer-linear maps in (3). Nonnegativity is preserved in both directions. A coordinate repetition neither changes the gcd nor identifies different rays, establishing the lattice and extremal claims. $\square$

When $k = 0$, every global vertex class contracts to a singleton. The full normal cone is generated by vertex links, and there is no disc candidate. For $k > 0$, a non-link extreme ray is zero in each removed factor. Thus the kernel suffices to discover every non-link standard vertex surface in C_S .

Remark 3.3 (What this kernel does and does not compress). The theorem is a reduction of the algebraic matching-cone search. The ambient triangulation and the $O(t)$ expansion map are still needed to interpret a ray as a surface and check essentiality. We do not obtain a three-manifold with $O(k)$ tetrahedra, nor a full topological instance with $O(k)$ input bits. The implementation uses dense k -entry edge labels, so its straightforward preprocessing is polynomial in t and k , rather than a claimed strictly linear-time graph routine.

4 Primitive coordinate heights from incidence structure

4.1 A quick general determinant bound

For a nonnegative homogeneous cone $\{x \geq 0 : Ax = 0\}$, a positive vector with exact support J spans an extreme ray if and only if

$$\dim \ker A_J = 1. \quad (4)$$

If the nullity is larger, a nonproportional kernel perturbation preserves positivity for sufficiently small perturbations in both directions. If the nullity is one, any nonnegative decomposition stays on that support and hence on that one ray. This elementary criterion is also our exact extremality test.

Select $|J| - 1$ independent rows of A_J . Signed maximal minors give an integer nullvector, whose primitive version is obtained by changing an overall sign and dividing by its gcd. Applying Hadamard directly to the rows of M_S gives $\|x\|_\infty \leq 2^{4k}$, because their norms are at most two and their rank is at most $4k$. The following refinement uses more of the matrix structure and improves this to 4^k .

4.2 Only quadrilateral columns destroy total unimodularity

Write the reduced matrix as

$$M_S = [B \mid D], \quad (5)$$

where B contains the triangle-class columns and D the allowed quadrilateral columns. The rows of B are directed graph incidences: an entry -1 at the source class, $+1$ at the target class, and zeros elsewhere. A loop gives a zero row. Thus B is a transpose of an incidence matrix, with some columns possibly absent.

Lemma 4.1 (Incidence minors). *Every square minor of B has determinant in $\{-1, 0, 1\}$.*

Proof. Induct on the size of the square submatrix. A row with zero or one nonzero entry permits a determinant expansion and the induction hypothesis. Otherwise every row retains both of its opposite entries, so the all-ones vector is in the kernel and the determinant is zero. The empty determinant is one. $\square$

Set

$$\delta_j = \|D_j\|_1 \leq 4, \quad \Delta_j = \max(1, \delta_j).$$

Lemma 4.2 (Mixed minors). *For any square submatrix A of $[B \mid D]$, with chosen quadrilateral columns indexed by I ,*

$$|\det A| \leq \prod_{j \in I} \|D_j|_{\text{rows}(A)}\|_1 \leq \prod_{j \in I} \delta_j. \quad (6)$$

Proof. Expand successively along the chosen quadrilateral columns. Each surviving term chooses distinct rows and finishes with a minor of B , of absolute value at most one. Bounding the sum of absolute terms by all row choices, including choices which would repeat a row and thus cannot occur, gives the product of column absolute sums. If a selected column is zero, every corresponding minor vanishes, so the formula remains valid. $\square$

Theorem 4.3 (Support-dependent primitive height). *Let x be a primitive non-link standard extreme ray in the sector S , and let I be its actual positive quadrilateral support, of size $a \leq k$. Then every triangle coordinate and every positive quadrilateral coordinate obey*

$$a_c \leq \prod_{j \in I} \Delta_j \leq 4^a, \quad (7)$$

$$q_i \leq \prod_{j \in I \setminus \{i\}} \Delta_j \leq 4^{a-1} \quad (i \in I). \quad (8)$$

In particular $\|x\|_\infty \leq 4^k$, and every coordinate needs at most $2k + 1$ binary digits.

Proof. Apply (4) to the exact positive coordinate support J in the reduced kernel. Choose $|J| - 1$ independent rows. A maximal minor omitting a triangle column contains all quadrilateral columns in I ; a minor omitting quadrilateral column i contains only $I \setminus \{i\}$. The mixed-minor bound gives the respective products. Gcd division can only decrease absolute coordinates. Expansion to the full triangulation merely repeats triangle coordinates and inserts zeros, so the bounds hold in the original vector.

For completeness, if an active quadrilateral column has $\delta_i = 0$, it is an independent free coordinate of the cone. The only extreme ray using it is its coordinate axis, with $q_i = 1$ and all other coordinates zero. Replacing δ_i by Δ_i makes the displayed bound valid without a zero-column exception. If $a = 0$, the ray is a vertex link and has only zeros and ones. $\square$

This proof locates the source of large coordinates precisely: the triangle incidence block has unit minors, and only the quadrilateral columns contribute multiplicative growth. It is a support-sensitive use of the classical cofactor method, not a new general determinant principle.

4.3 A consequence for fundamental surfaces

The same bound extends, with a factor of the support size, to indecomposable integer normal vectors. This consequence is useful for future searches, although the delivered code enumerates vertex rays only.

Corollary 4.4 (Fundamental-coordinate bound). *Let x be a non-link fundamental normal vector, meaning that it is not a sum of two nonzero admissible integral vectors. If its actual quadrilateral support has size $a \geq 1$, then*

$$\|x\|_\infty \leq a4^a, \quad q_i \leq a4^{a-1}. \quad (9)$$

If x is not itself a primitive vertex ray, both bounds are strict.

Proof. Indecomposability forces x to be canonical: otherwise an integral vertex link could be subtracted, giving a nontrivial decomposition. The minimal face containing x therefore has at least one zero triangle anchor at each global vertex. Quadrilateral projection is injective on the

linear span of that face. Indeed, a matching vector with zero quadrilaterals is a linear combination of vertex links, and the zero anchors force every such coefficient to vanish. The face consequently has dimension at most a .

Conical Carathéodory gives $x = \sum_{j=1}^b \lambda_j r_j$ with $b \leq a$, $\lambda_j \geq 0$, and primitive extreme rays r_j of that face. Their actual quadrilateral supports are contained in that of x . If some $\lambda_j \geq 1$, the integral vector $x - r_j$ is nonnegative and matching. It is admissible in the same sector, so fundamentality forces $x = r_j$. Otherwise every coefficient is less than one. Applying Theorem 4.3 to at most a rays gives the strict bounds. The vertex-ray case gives the weak bounds directly. $\square$

4.4 Consequences for bit complexity and supplied-vector reduction

The normal-disc count of a primitive standard ray is bounded by

$$\|x\|_1 \leq 7t 4^k. \quad (10)$$

An edge intersection is the sum of the two incident triangle coordinates and at most one allowed quadrilateral coordinate, so it is at most $3 \cdot 4^k$. Thus the logarithm of every interval endpoint needed by the compressed component routines is $O(k + \log t)$. The theorem is consequently meaningful in bit complexity: it does not silently charge unit cost for enormous normal surface weights.

The height theorem applies to primitive extreme rays. It does not bound an arbitrary supplied surface, which may have unbounded multiplicity or be a large sum of rays. For arbitrary canonical surfaces, a separate triangle range bound is useful.

Proposition 4.5 (Triangle range). *For any nonnegative matching quadrilateral vector in the sector, the range of triangle values at each global vertex satisfies*

$$\max a_c - \min a_c \leq 4 \sum_{j=1}^k q_j \leq 4k \max_j q_j \quad (k \geq 1). \quad (11)$$

Proof. Compare a maximum and minimum corner along a simple path in the triangle graph after zero-edge contraction. Sum the edge differences. Every allowed quadrilateral has at most four occurrences in the entire graph, so its total absolute contribution to the path is at most $4q_j$. The same argument works for a path traversing some edges backwards. Adding a vertex link shifts all values at that vertex equally and leaves the range unchanged. $\square$

For a supplied vector with quadrilateral gcd $g > 0$, subtract vertex links and divide by g as in the maintained disc-count core. If its actual support is a , its resulting triangle bound can be sharpened from the earlier $(4t - 1)Q_{\max}/g$ estimate to $4aQ_{\max}/g$ whenever the latter is smaller. This is a proved estimate; this package does not alter the existing core routine, whose arithmetic already constructs the exact reduced vector.

5 Canonical lifts and the positive-Euler preliminary search

5.1 Integrating the difference equations

Choose a root and spanning tree in each contracted triangle component. Integrating (1) along the tree gives integer linear forms $\ell_c(q)$ with root value zero. For every remaining edge impose

$$\ell_d(q) - \ell_c(q) = b_e(q). \quad (12)$$

Collect the residual equations in a matrix R_S . Then the quadrilateral cone is exactly

$$Q_S = \{q \in \mathbb{R}^k : q \geq 0, R_S q = 0\}. \quad (13)$$

Indeed, consistent potentials determine matching triangles up to one additive constant per global vertex, and choosing sufficiently large constants always makes the triangles nonnegative.

Each coefficient of a potential, a within-vertex difference of potentials, or a fundamental-cycle residual has absolute value at most four. A root path is simple; when two root paths are subtracted their common segment cancels; a fundamental cycle uses every relevant edge at most once. Each variable still has only four original occurrences. This observation gives polynomial bit sizes for the exact linear algebra below.

Let $d = \dim \ker R_S$ and $r = k - d$. This is the dimension of the linear matching space. The nonnegative cone may have smaller linear span; our algorithms remain correct in that degenerate case, although their bounds can be loose.

5.2 Canonical extension

The canonical lift $F(q)$ subtracts the minimum potential at every vertex:

$$F(q)_c = \ell_c(q) - \min_{w \text{ at } v(c)} \ell_w(q), \quad \pi_Q F(q) = q. \quad (14)$$

It has a zero triangle at each global vertex and contains no vertex-link component. Every other nonnegative lift is uniquely

$$F(q) + \sum_v \lambda_v L_v, \quad \lambda_v \geq 0. \quad (15)$$

All potentials are integral linear forms, so $F(q)$ is integral for integral q . Moreover q is primitive exactly when $F(q)$ is primitive: every common divisor of q divides all canonical triangle entries, and the quadrilaterals are coordinates of the full vector. These are the canonical-extension properties of Q-theory [1, 2].

Lemma 5.1 (A canonical Q-ray is standard extreme). *If q spans an extreme ray of Q_S , then $F(q)$ spans an extreme ray of C_S .*

Proof. Suppose $F(q) = y + z$ in C_S . Projection to quadrilateral coordinates makes $\pi_Q y$ and $\pi_Q z$ nonnegative multiples of q . By (15), y and z are the corresponding multiples of $F(q)$ plus nonnegative sums of vertex links. But $F(q)$ has a zero triangle at every global vertex. At those corners the link coefficients in $y + z$ must sum to zero, so all vanish. Thus both summands lie on the original ray. $\square$

This is the sector-restricted form of Burton's Lemma 4.5 [2]. The converse is false in general: standard vertices need not project to Q-vertices, which is why a second enumeration phase is retained.

5.3 Convexity gives an exact positive-Euler decision

Define $g(q) = \chi(F(q))$. Euler characteristic is linear on the standard matching space, as can be seen by assigning normal cells to their ambient edges, faces, and tetrahedra. Consequently

$$g(q) = \lambda(q) - \sum_v \chi(L_v) \min_{c \text{ at } v} \ell_c(q) \quad (16)$$

for a linear form λ . Since the minimum of linear functions is concave and every coefficient $\chi(L_v)$ is positive, g is convex and positively homogeneous. Equivalently,

$$g(q + q') \leq g(q) + g(q'), \quad g(\alpha q) = \alpha g(q) \quad (\alpha \geq 0). \quad (17)$$

The same proof appears for canonical Euler characteristic in Burton’s almost-normal setting [3, proof of Theorem 5.5]; here it is applied to compact support-restricted normal cones, with disc as well as sphere vertex links.

Theorem 5.2 (Q-ray screening is complete for positive canonical Euler). *The sector contains a canonical normal surface of positive Euler characteristic if and only if a primitive extreme ray of Q_S has positive Euler characteristic after canonical lifting.*

Proof. The pointed rational polyhedral cone Q_S is generated by its finitely many primitive extreme rays r_i . Write $q = \sum_i \alpha_i r_i$, with $\alpha_i \geq 0$. By (17),

$$0 < g(q) \leq \sum_i \alpha_i g(r_i)$$

forces at least one positive $g(r_i)$. The converse takes $q = r_i$. By Lemma 5.1, each primitive canonical ray is a connected non-link standard vertex surface, so this is a finite search for actual surfaces and not just a relaxation with fractional answers. $\square$

Proposition 5.3 (Rank-sensitive first phase). *If $d > 0$, the complete Q-ray phase attempts at most*

$$\binom{k}{d-1} = \binom{k}{r+1} \quad (18)$$

homogeneous linear systems, with polynomial bit work per system. If $d = 0$ there are no nonzero candidates.

Proof. An extreme ray of Q_S is cut out inside $\ker R_S$ by coordinate hyperplanes whose restrictions have rank $d - 1$. Choose any independent $d - 1$ of them. Enumerating all such subsets, retaining exactly one-dimensional intersections and their nonnegative orientation, covers all rays. Degenerate rays may have additional zero coordinates and therefore appear more than once; primitive normalization and exact deduplication remove these repetitions. The implementation first projects and deduplicates hyperplanes in a rational kernel basis, which can reduce the number further. $\square$

Thus a supplied dense sector can be easy when d is small. A small matching rank r is also useful for this preliminary phase because of the complementary binomial expression. Neither fact solves the separate problem of choosing a useful compatible sector.

5.4 Positive Euler is not essentiality

A connected positive-Euler normal surface can be a sphere, a projective plane, or a disc whose boundary is inessential. The projective-plane possibility is not excluded merely by orientability of the ambient three-manifold. Even irreducibility does not exclude boundary-parallel normal discs which are not vertex links. Burton–Ozlen’s complete recognition procedure handles such surfaces through crushing and iteration [5]; this module does not silently invoke such an unimplemented transition.

Our independent corpus contains a particularly small guard. Attach one tetrahedron along a boundary face of the one-tetrahedron layered solid torus. The resulting two-tetrahedron solid torus has canonical Q-ray discs with inessential boundary. Their sectors have dimension one and positive Euler characteristic, but the disc checker correctly rejects essentiality. The corpus also includes a solid torus connected-summed with $\mathbb{R}P^3$, where a canonical Q-ray is a projective plane. These are direct tests of the logical distinction, rather than artificial malformed inputs.

6 Recovering every standard ray in the sector

6.1 Exact positive-support enumeration

Let $N \leq 9k$ be the kernel variable count and $s = \text{rank } M_S \leq 4k$. For each coordinate subset J with $1 \leq |J| \leq s+1$, test (4). A one-dimensional nullspace with a vector strictly positive on J gives exactly one primitive ray with that exact support. Skip subsets using no quadrilateral variable, since they give vertex links. This enumerates every non-link standard ray without duplication and uses at most $2^N \leq 2^{9k}$ linear systems.

The upper limit $s+1$ follows from $\text{rank}(M_S)_J = |J| - 1$. The actual number of attempted supports is

$$B_S = \sum_{j=1}^{\min(N, s+1)} \left[\binom{N}{j} - \binom{p}{j} \right], \quad (19)$$

where p is the number of triangle-class variables and $\binom{p}{j} = 0$ for $j > p$. This exact count is used by the adaptive dispatcher.

6.2 Potential arrangements for small nullity

Work in $E = \ker R_S$, of dimension d . Include the coordinate hyperplanes $q_i = 0$ and, for every pair of retained triangle classes at the same global vertex, the hyperplane

$$\ell_c(q) - \ell_{c'}(q) = 0. \quad (20)$$

Delete forms that vanish identically on E , and deduplicate proportional forms exactly. If there are H remaining hyperplanes, then for $k \geq 1$

$$H \leq k + \binom{8k}{2} \leq 32k^2. \quad (21)$$

On every closed chamber of this central arrangement inside Q_S , the minimum potential at each global vertex is fixed, so the canonical lift F is linear.

Theorem 6.1 (Arrangement coverage). *Every non-link standard extreme ray in C_S is the canonical lift of a nonnegative one-dimensional intersection of E with $d-1$ independent hyperplanes from (20) and the coordinate hyperplanes.*

Proof. Let x be such a standard ray and $q = \pi_Q x$. A non-link standard extreme ray is canonical: a positive vertex-link summand in (15) would give a nonproportional decomposition. Choose a closed arrangement chamber containing q . If q were not extreme there, a nonproportional decomposition $q = q_1 + q_2$ in that chamber would lift linearly to $x = F(q_1) + F(q_2)$, contradicting standard extremality. Hence q is a chamber ray and is cut out by $d-1$ independent boundary hyperplanes in E . $\square$

Some arrangement rays arise from a tie between potentials that are not minimal. They may be subdivisions with no corresponding standard extreme ray. Accordingly every lifted candidate undergoes the exact active-column rank test (4); we do not present the unfiltered arrangement as the standard vertex set. Only retained rays are subjected to the strong 4^k height assertion. Before this filter, a generic cofactor estimate gives $2^{O(k \log k)}$ coordinate magnitudes, still polynomial bit lengths for the per-candidate arithmetic.

Corollary 6.2 (Implemented adaptive enumeration). *For $d > 0$, all non-link standard rays in a specified sector can be enumerated in*

$$\text{poly}(t, k) \min \left\{ 2^{9k}, \binom{H}{d-1} \right\} \quad (22)$$

bit operations, up to polynomial factors accounting for exact arithmetic and output. The implementation compares (19) with the actual arrangement binomial and chooses the smaller method. The $d = 0$ case is empty; for $d = 1$ the arrangement tries one system.

6.3 Algorithmic summary

Listing 1: Fixed-sector discovery; all arithmetic is exact.

```

validate the finite triangulation and compatible allowed types
contract triangle edges with identically zero quadrilateral label
retain all nonzero-label equations, including loops
factor off free common triangle values (vertex links)
construct cycle equations and an exact quadrilateral kernel basis

enumerate Q-rays and canonically lift them
for each positive-Euler ray:
    construct the compressed essential-disc count certificate
    replay it independently
    if the count is positive: return DISC_FOUND
if every Q-ray has nonpositive Euler: return NO_POSITIVE_EULER

if complete standard search is requested:
    choose kernel-support or potential-arrangement enumeration
    filter every candidate by exact standard extremality
    replay essential-disc certificates for positive-Euler rays
    return the first verified disc, or NO_VERTEX_DISC_IN_SECTOR
otherwise return POSITIVE_EULER_ONLY

```

The polynomial coefficient-arithmetic factor is intentionally not represented as a small fixed exponent for this Python prototype. Rational elimination, gcd normalization, source validation, and integer serialization all contribute. The theorem concerns the dependence on the combinatorial search parameters.

7 From a ray to an essential-disc certificate

7.1 The maintained weighted normal-component checker

The existing normal-surface component code constructs an interval-pairing system with one orbit for each connected surface component. It transports three integer weights: Euler characteristic and two boundary intersection counts representing a homology basis of the torus. Weighted Agol–Hass–Thurston orbit operations are polynomial in the encoded input, including logarithmic endpoint size [8, Theorems 12 and 16, Corollary 17]. The project supplies an independent weighted trace verifier and exact source binding.

For a connected compact surface component C , the conditions

$$\chi(C) = 1, \quad [\partial C] \neq 0 \quad \text{in } H_1(\partial M; \mathbb{F}_2) \quad (23)$$

are equivalent to being a properly embedded disc with essential boundary on the torus. Nonzero boundary class implies that C has boundary. Surface classification then forces C to be a disc. An

essential embedded circle on a torus has primitive integral slope and hence nonzero reduction modulo two. A closed projective plane has zero boundary class and is excluded.

The checks are componentwise. Aggregate parity can cancel for two parallel discs, and aggregate Euler characteristic can obscure other components. Although our primitive standard rays are mathematically connected, the implementation deliberately reuses the full established component checker. This tests the geometry and encoded surface independently of the new extreme-ray argument.

7.2 Positive witness schema

A returned `normal-sector-witness-v1` certificate contains the exact triangulation digest, sorted allowed types, and the full reconstructed normal coordinate vector. It also includes the existing disc-count certificate, with schema `normal-disc-count-v1`. The new witness verifier independently checks the source, the allowed support, all normal equations, and a strictly positive checked disc count. It does not need to trust how the ray was discovered. In particular it does not import the producer's contraction or enumeration routines.

This is a strong asymmetry: verifying a supplied positive witness uses polynomial encoded operations, while discovering that witness may take exponential time over sectors. The new package preserves that asymmetry rather than inferring a fast search from a fast checker.

7.3 Exhaustion replay uses an independent dense route

A complete sector certificate records the phase, the complete primitive quadrilateral-ray list, each exact Euler value, and a checked zero-disc certificate for every positive-Euler ray which was rejected. Its verifier reconstructs the answer by a separate path:

1. Build the original dense standard matching matrix from the validated face equations, retaining $4t$ triangle columns and the allowed q columns.
2. Eliminate triangle columns by rational row reduction. Rows with zero triangle part give independent quadrilateral consistency equations.
3. Set free triangle variables to zero to obtain a separate affine lift, then subtract minima at each global vertex to make it canonical.
4. Reconstruct the Q-ray or arrangement ray set using separate exact linear-algebra code; filter standard extremality in the original matrix.
5. Compare the complete set, Euler values, and native negative disc certificates, then derive the exact permitted status.

The two modules share geometry validation and the older disc checker. They do not share their elimination, potential construction, contraction, or ray enumeration helpers. The audit disables every producer-defined function while running the new verifiers.

Exhaustion replay repeats a search. It is *not* claimed polynomial in the length of a short submitted negative transcript: the verifier must establish that no ray was omitted, and repeated or low-output instances can require many linear systems. Its relevance is independent correctness and the same parameter-restricted complexity regime. A future Farkas-style negative certificate is a separate research goal.

7.4 Resource and source semantics

The basis allowance counts attempted linear systems, including unsuccessful ones, rather than only successful rays. This prevents a purported ray limit from hiding an unbounded sequence of rejected candidates. An incomplete sector query returns `INCONCLUSIVE` without an exhaustion certificate. The optional native orbit allowance is per positive candidate. In the outer driver,

Q and standard phases each receive their explicitly named per-sector basis allowance; there can therefore be two such stages for one sector. A shared cooperative callback can impose a global time or cancellation condition across phases and across certificate replay.

Callbacks are called through geometry validation, contraction, exact elimination, candidate enumeration, component checks, and independent replay. False-valued callable objects are still invoked. The audit checks that a callback raising `ValueError` inside either verifier propagates as cancellation rather than being converted into a rejected certificate.

The byte digest binds the exact labelled triangulation. A relabelling that represents an isomorphic manifold is still a different source and requires a new certificate or an explicitly certified relabelling. This is intentional source binding; it is not an isomorphism-invariant manifold identity.

8 Outer support search and conditional complexity

Definition 8.1 (Vertex-disc support parameter). Let $\kappa(\mathcal{T})$ be the minimum number of tetrahedra with a nonzero quadrilateral among all essential *standard vertex* discs in $\mathcal{T}$, with $\kappa(\mathcal{T}) = \infty$ if no such disc exists.

The outer search enumerates choices of j tetrahedra and one of three types in each, for $j = 1, \dots, K$. It first runs the complete Q phase. A checked disc terminates immediately. A negative positive-Euler result excludes that sector. Otherwise it enumerates all standard rays there. Rays belonging to several sectors may be encountered repeatedly; the upper bound includes those repetitions.

Theorem 8.2 (Support-limited discovery). *With resource caps disabled, this search finds an independently checked essential-disc witness whenever $\kappa(\mathcal{T}) \leq K$. Its running time is bounded by*

$$\text{poly}(t, K) \left(1 + \sum_{j=1}^{\min(K, t)} \binom{t}{j} 3^j 2^{9j} \right). \quad (24)$$

For $K = O(\log t)$ this is $2^{O((\log t)^2)} = t^{O(\log t)}$. For a known sector the search is fixed-parameter tractable in k ; searching for an unknown support by (24) is an XP algorithm in K .

Proof. Every vertex disc of support at most K belongs to one enumerated compatible sector. Q-phase rejection is safe by Theorem 5.2; if that phase does not decide the sector, standard enumeration includes its primitive ray. The essential-disc check then succeeds. There are $\binom{t}{j} 3^j$ sectors of size j , and Corollary 6.2 gives the stated safe per-sector bound. The coordinate-height theorem and (10) make component verification polynomial in t and j . For logarithmic K , t^K and all remaining exponential-in- K factors fit the displayed quasi-polynomial estimate. $\square$

The global vertex-disc theorem of Jaco–Tollefson, stated explicitly as Theorem 5.6 by Hass–Lagarias–Pippenger [6, 7], guarantees an essential standard vertex disc whenever a compact triangulated three-manifold with nonempty boundary has an essential disc. Therefore $K = t$ yields complete boundary-compressibility search. With verified knot-exterior provenance this would be a complete unknot decision procedure, but (24) is exponential when $K = t$. Its coarse exponential base is not advertised as an improvement over optimized general normal-surface enumerators.

For $K < t$, exhaustion has only the support-qualified meaning in Table 1. The global vertex-disc theorem does not, by itself, say that an arbitrary normal disc in a particular coordinate face can be replaced by an essential vertex disc in that same face. We therefore do not replace κ by the minimum support of an arbitrary normal disc. That stronger face-preserving statement would require a separate proof.

There are two useful conditional quasi-polynomial regimes:

- a class of triangulations for which positive instances satisfy $\kappa(\mathcal{T}) = O(\log t)$, with a controlled construction from diagrams;
- a quasi-polynomially generated family of compatible sectors containing an essential vertex disc, with sufficiently small d for complete standard enumeration, or small $\min(d-1, r+1)$ when the preliminary Q phase itself finds the disc.

The prototype supplies the fixed-sector algorithms. It proves neither a universal small-support theorem nor an efficient global generator of the required sectors.

9 A rigorous obstruction to uniformly sparse meridians

Let $F_0 = 0$, $F_1 = F_2 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Burton–Paixão–Spreer construct the one-vertex t -tetrahedron layered solid tori

$$B_t = \text{LST}(F_{t+1}, F_{t+2}, F_{t+3}). \quad (25)$$

The three boundary edges have the indicated meridional intersection numbers, and their Theorem 3.3 exhibits a vertex meridian disc with maximum normal coordinate F_{t+1} [9].

The existence of one large disc would not rule out a different sparse one. The height theorem permits the following stronger deduction.

Theorem 9.1 (Every vertex meridian needs linear support). *For the triangulations B_t in (25), every essential standard vertex disc with quadrilateral support satisfies*

$$a \geq \frac{\log_2 F_{t+3} - \log_2 3}{2}. \quad (26)$$

Consequently $\kappa(B_t) = \Omega(t)$.

Proof. Every essential disc boundary in a solid torus has meridian slope. On the boundary edge whose minimal meridional intersection is F_{t+3} , the boundary of any such disc has at least F_{t+3} intersections. Normal detours can increase this count, but cannot decrease the geometric intersection number of the slopes.

The coordinate vector of a normal disc is primitive: otherwise dividing it by an integer $g > 1$ would give an integral admissible normal vector with Euler characteristic $1/g$, impossible. Thus Theorem 4.3 applies to a standard vertex disc. Choose any tetrahedral representative of the relevant boundary edge. Its intersection weight is the sum of its two endpoint triangle coordinates and at most one active quadrilateral coordinate, so it is at most $3 \cdot 4^a$. Hence

$$F_{t+3} \leq 3 \cdot 4^a,$$

which gives (26). Finally F_{t+3} grows exponentially in t , so its binary logarithm is linear in t . $\square$

This lower bound is derived here from the classical layered-torus construction and the support-height theorem. It is a statement about the chosen triangulation, not a lower bound on unknot-recognition time. All B_t are the same simple manifold topologically. A different triangulation or a compressed algebraic description may make their discs easy to find.

Corollary 4.4 gives the same asymptotic obstruction for essential fundamental meridians: their support a satisfies $F_{t+3} \leq 3a4^a$, and therefore $2a + \log_2 a \geq \log_2 F_{t+3} - \log_2 3$. Hence $a = \Omega(t)$ there as well. Replacing vertex enumeration by Hilbert-basis enumeration alone cannot restore a universal sparse-meridian theorem for arbitrary solid-torus triangulations.

9.1 Dense support can coexist with a one-dimensional search

The reproducible family in the package gives exactly such a contrast. Number tetrahedra $0, \dots, t-1$ and allow quadrilateral type 2 in every tetrahedron (the implementation's zero-based convention). In the explicit labelling of `scripts/fibonacci_fixture.py`, the primitive meridian rows are

$$(F_{t-i+1}, F_{t-i+1}, 0, 0, 0, F_{t-i}), \quad i = 0, \dots, t-1. \quad (27)$$

The two exposed boundary faces are in tetrahedron zero. The face maps and formula are independently checked against Regina's layered-solid-torus construction for $t = 1, 2, 3, 4, 8, 12, 16, 24, 32$.

The full allowed sector has $k = t$ but $d = 1$. The Q-ray phase therefore tries one direction and recovers exponentially large coordinates without enumerating their individual normal discs. The total number of normal discs in (27) is

$$\sum_{i=0}^{t-1} (2F_{t-i+1} + F_{t-i}) = F_{t+5} - 5. \quad (28)$$

These examples motivate separating deterministic propagation from combinatorial choices of allowed types. They do not show that the correct dense sector can always be selected efficiently.

The dimension-one assertion holds for the whole family, not merely for the tested sizes. To see it directly, write the triangle coordinates in tetrahedron i as (A_i, B_i, C_i, D_i) and its one allowed quadrilateral as q_i . The explicit internal face maps in the generator give

$$\begin{aligned} A_i &= A_{i+1} + q_{i+1}, & B_i &= B_{i+1} + q_{i+1}, \\ C_i &= D_{i+1}, & D_i &= C_{i+1}, \\ D_i + q_i &= A_{i+1}, & C_i + q_i &= B_{i+1}. \end{aligned}$$

The final tetrahedron's self-gluing gives $A = B = C + q$ and $C = D$. Putting $u = q_{t-1}$ and $s = C_{t-1}$, backwards induction therefore yields

$$C_i = D_i = s, \quad A_i = B_i = s + F_{t-i+1}u, \quad q_i = F_{t-i}u.$$

The standard cone is exactly the two-parameter cone $s, u \geq 0$: s is its vertex-link coordinate and u its meridional coordinate. Canonical lifting sets $s = 0$, proving both $d = 1$ and (27) uniformly in t . The finite Regina comparisons test the implementation of these formulae; they are not being used as a proof of the infinite recurrence.

10 Validation and reproducible measurements

10.1 Independent oracle construction

The frozen corpus was constructed with Regina 7.4 [12]. It contains 48 compact orientable triangulations with one torus boundary: Fibonacci and other layered solid tori, tetrahedra attached along boundary faces, interior 1–4 moves, legal 2–3 moves, compact trefoil and figure-eight exteriors, and two connected-sum essentiality guards. It includes multiple boundary vertices, interior vertices, loops, and self-gluing.

Complete standard and Q vertex lists were enumerated independently by Regina. The corpus contains 1,801 standard vertices and 617 Q vertices. A second script filters these already complete lists by allowed sectors and computes the sector nullity from Regina's Q-matching matrix using independent exact elimination. No expected vector is obtained from the new producer.

All 4^t partial type assignments are used on the ten fixtures with $t \leq 3$, giving 388 exhaustive sectors. Larger cases include occupied vertex supports of nullity at most two, full meridian sectors,

selected nullity-three sectors, and empty controls. The complete selection has 1,718 sectors. This selection is deliberate coverage of the mathematical mechanisms, not a random sample from a distribution of knot diagrams.

10.2 Exact comparisons and adversarial replay

The following counts are completed tests, not projected test plans.

Audit	Completed comparisons
Standard-ray set equality with Regina, all selected sectors	1,718
Q-ray canonical-lift set equality with Regina	1,718
Standard ray occurrences compared	3,249
Q-ray occurrences compared	2,361
Forced kernel-support enumeration, $k \leq 2$	502
Ray occurrences in forced-support audit	532
Complete discovery certificates independently replayed	962
False-assertion mutations rejected	60
Explicit allowance checks	3
False-valued callback propagation checks	4
Focused public-contract tests	17
Self-contained repository integration tests	8

Table 2: All listed comparisons passed; no selected enumeration was capped. Raw per-case data and source hashes accompany the package.

The 962 discovery replays use 481 sectors in both phases: all 388 exhaustive small sectors and 93 larger or essentiality-guard sectors. They yield 694 NO_POSITIVE_EULER, 134 DISC_FOUND, 67 POSITIVE_EULER_ONLY, and 67 NO_VERTEX_DISC_IN_SECTOR results. Every producer-defined function is disabled during independent replay. Mutations change the source gluing, digest, schema, allowed support, witness coordinates, claimed disc count, required proof, ray completeness, ray uniqueness, Euler value, quadrilateral direction, and claimed status. All false assertions are rejected.

The separate oracle-selection file has 400 sectors with positive canonical Euler characteristic but no essential standard vertex disc. These cases directly exercise the distinction on which the two-phase interface relies. The full set equality tests also detect artificial arrangement rays: only standard-extremality-filtered rays may be emitted.

The additive patch was applied to a temporary copy of the retrieved baseline. Its eight self-contained tests passed there and in the standalone package, with Python site packages disabled. Patch applicability, exact new-file contents, reverse applicability, and unchanged baseline source hashes were checked. These integration tests require neither Regina nor the frozen oracle files.

10.3 Benchmark design

There are two distinct timing questions, and the data preserve that distinction.

1. *The same exact fixed-sector Q-ray task.* We compare the new contracted implementation with the independent dense Python reference. Both validate the same finite triangulation, enumerate all canonical Q-ray lifts for the supplied sector, and produce exactly the same vector set. Neither timing includes essential-disc verification. This measures the computational effect of eliminating redundant triangle unknowns; the dense implementation is a transparent reference, not a claim about the best existing library.
2. *A complete new certified discovery call.* On dense Fibonacci sectors, we measure discovery, native disc-certificate construction, independent native replay, and certificate serialization. The

supplied sector is known in advance. No ratio to a whole recognizer or to a method without that sector information is reported.

For the matched comparison, the family starts from a one-tetrahedron solid torus and repeatedly attaches a tetrahedron along a boundary face. Only the original meridional quadrilateral type is allowed, so $k = 1$ and $d = 1$. For all measured sizes the new kernel has six variables and four equations, while the dense reference starts with $4t + 1$ variables. Fixture construction is outside the clock.

Four shuffled arms, two identical A/A copies of each implementation, run one retained warm-up and five measured rounds. Ratios are formed inside a matched round before taking their median. For Fibonacci certified calls there are two shuffled copies of the same new method, one warm-up per copy, and five rounds through $t = 64$; the $t = 128$ case uses three rounds. All raw calls, warm-ups, orderings, exact fixtures, answer hashes, certificate sizes, and source hashes are retained.

t	Dense variables	Kernel variables	Kernel (ms)	Dense (ms)	Paired ratio
4	17	6	0.338	1.430	4.24
8	33	6	0.580	6.569	11.05
16	65	6	1.567	36.457	23.55
32	129	6	1.864	104.015	58.70
64	257	6	3.859	400.672	108.13
128	513	6	7.391	1,850.127	251.63

Table 3: The identical fixed-sector Q-ray task on boundary-cap triangulations. Every case has $k = d = 1$. Times are medians of ten measured calls per method; the last column is the median of ten within-round dense/kernel ratios. The comparator is the independent dense Python reference.

$t = k$	Maximum coordinate bits	Certificate bytes	Median (s)	Measured calls
4	3	4,747	0.006032	10
8	6	10,700	0.010546	10
16	11	32,752	0.042811	10
32	22	117,706	0.226509	10
64	44	455,055	1.542706	10
128	89	1,792,597	11.075735	6

Table 4: Complete certified discovery on supplied dense Fibonacci sectors. All have $d = 1$ and return one verified essential meridian. The clock includes native independent disc replay and JSON serialization. These are timings of the new method, without a cross-algorithm ratio.

The main run completed 176 measured calls and 36 retained warm-up calls on Python 3.12.14. The frozen runtime and benchmark source hashes were unchanged from the beginning to the end of that run. A first exploratory timing run is also retained separately; the tables use only the second, controlled run. No other agent computation was scheduled concurrently with this main run.

10.4 What the measurements establish

The sparse family exhibits the intended removal of triangle-only propagation from the expensive algebra. The dense Fibonacci family shows that large support and huge coordinate magnitudes can be handled by the dimension-one search without sheet expansion. In both cases the measured task remains a given-sector query.

The corpus also records Regina’s default, direct standard double-description, and via-reduced tree enumeration times, with agreement of their complete standard vector sets. These are separate reference measurements. Regina’s compiled enumerators are very fast on the small fixtures;

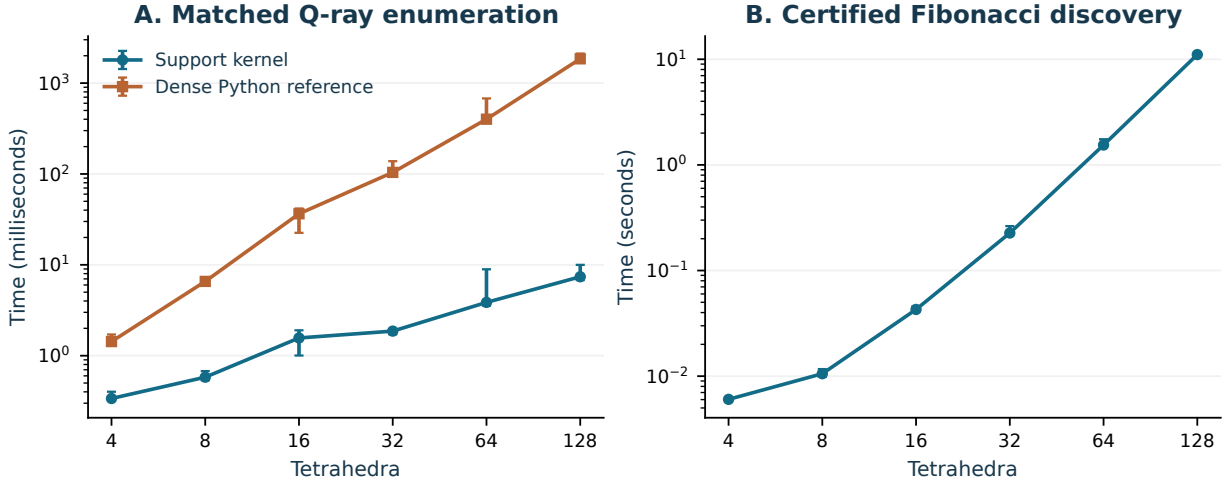


Figure 1: Measured scaling for two different supplied-sector workloads. Points are medians; bars span the observed minimum and maximum. Both axes are logarithmic. The bars describe repeat variability, not confidence intervals or a distribution over inputs.

our Python code is not claimed faster than Regina’s optimized implementation. Nor do the measurements establish a speedup in the maintained end-to-end knot pipeline, which does not yet call this new sector search.

The timing data come from one execution environment and repeated identical fixtures. They describe these workloads; they are not evidence of a new unconditional asymptotic bound beyond the proved parameterized results.

11 Integration, reproducibility, and practical use

11.1 Package layout and dependencies

The archive contains the complete article source and PDF, the two additive runtime modules, an unchanged upstream runtime snapshot, deterministic fixture generators, the frozen independent oracle, audit scripts, all raw results, focused tests, an integration patch, and source manifests. The runtime search uses the Python standard library only. Regina 7.4 is needed only to regenerate the external oracle and its optional reference timing measurements. LaTeX and Poppler are used for the report build and visual inspection.

The two new modules can be copied into the existing `Topology/UnknotRecognition/fast/fastunknot/` directory. The included `integration.patch` uses repository-relative paths. The existing recognizer is not modified to treat these supplied-triangulation results as diagram verdicts. Its normal fallback and its external-backend provenance remain separate interfaces.

11.2 Minimal API example

From the archive root:

```
import sys
sys.path[:0] = ["fast", "scripts"]
from fibonacci_fixture import fibonacci_torus
from fastunknot.normal_sector import discover_in_sector
from fastunknot.normal_sector_verify import verify_sector_witness

fixture = fibonacci_torus(16)
triangulation = fixture["triangulation"]
allowed = [(i, 2) for i in range(16)]
result = discover_in_sector(triangulation, allowed)
assert result["status"] == "DISC_FOUND"
assert verify_sector_witness(triangulation, result["certificate"])
```

For a full sector list, use `enumerate_sector` with `phase='standard'` or `phase='quadrilateral'`. Standard enumeration accepts `method='auto'`, `'arrangement'`, or `'supports'`. The auto method realizes the dispatch in Corollary 6.2.

For bounded outer search:

```
from fastunknot.normal_sector import sparse_disc_search
result = sparse_disc_search(triangulation, max_active=2,
                           max_sectors=1000,
                           max_bases_per_sector=10000)
# Inspect status. A support cutoff is never a knot verdict.
```

11.3 Reproduction commands

The README gives complete commands with explicit portable paths. The main entry point runs the frozen-oracle checks without requiring Regina:

```
python3 reproduce.py --quick
python3 reproduce.py --audits
python3 scripts/benchmark.py --large --output results/benchmarks_rerun.json
python3 reproduce.py --build-pdf
```

Regenerating the oracle is a separate operation because changing the Regina version or simplification choices can change the exact triangulation labelling. The frozen triangulations, their isomorphism signatures, and all expected vectors are sufficient to rerun the stated comparisons exactly.

11.4 Limits of the present implementation

The producer repeatedly prepares geometry when delegating to native component queries. A trusted immutable prepared-geometry object could reduce that overhead, but source binding would need to remain explicit. The dense exhaustion verifier deliberately favors a structurally different computation over maximum speed. The support search also revisits rays lying in multiple sectors and has no LP infeasibility cache, dynamic type propagation, or certified triangulation simplification. These are concrete optimization opportunities, not assumptions used to justify the current results.

12 Further research questions and proposed next steps

12.1 First priority: reduce choices while retaining dense propagation

The Fibonacci family shows that large support is sometimes unavoidable, but its correct sector has one-dimensional matching space. A useful next theorem should control the number of genuine choices of quadrilateral type, while allowing long deterministic propagation. Can a partial sector be completed by exact constraints so that only a small residual set of tetrahedra requires branching? A complete algorithm would need to prove that its forced choices are exhaustive and preserve an essential-disc possibility, not merely that they work on layered examples.

A concrete implementation project is to express forced chains as integer transfer matrices or short recurrences. One should track the remaining branching vertices and the bit length of composed maps. The hypothesis “small residual branching” must be checked on the actual search path. It cannot be replaced by the observation that some favorable sequence exists.

12.2 Second priority: useful negative certificates

Current complete sector replay reconstructs the ray set. Can absence of a positive canonical Euler surface be certified by a compact collection of linear dual witnesses, exploiting the incidence block and the structure of the minimum-potential function? A single linear Farkas certificate is not automatically enough for a convex piecewise-linear maximum. A successful construction must account for every relevant minimum chamber without encoding an exponentially long list when the output is small.

It is also worth seeking certifiable zero propagation: if exact feasibility forces a quadrilateral coordinate to vanish, a dual certificate could prune all descendant sectors sharing that partial assignment.

12.3 Third priority: certified changes of triangulation

Sparse-support promises are sensitive to the triangulation. Which elementary collapses, Pachner moves, layered-solid-torus reductions, or crushing moves can be applied with bounded encoded cost while preserving a checked correspondence to the original knot exterior? The desired output is not just a smaller triangulation; it includes maps sufficient to transfer a positive witness or a sound negative conclusion.

The main quantitative questions are whether such moves reduce support, reduce the dimension of useful sectors, or reduce the number of sectors needed. These objectives need not agree. A move that reduces tetrahedron count can increase the complexity of a preferred coordinate description.

12.4 Fourth priority: diagram-to-exterior provenance

Implement an independently replayable construction of the finite peripheral triangulation from the diagram, or a checked chain from an already trusted construction. This would let a successful `DISC_FOUND` become a source-bound unknot certificate. It would not, by itself, give a fast negative decision or a quasi-polynomial search bound, but it would connect the new positive search to the maintained recognizer soundly.

12.5 Fifth priority: essentiality within a coordinate face

Under what additional hypotheses does a face containing an essential normal disc also contain an essential standard vertex disc? The global Jaco–Tollefson theorem suffices for full search but does

not supply this stronger assertion as stated. Proving an appropriate support-preserving version would sharpen the parameter in Theorem 8.2 from vertex-disc support to arbitrary-disc support. A counterexample would be equally useful because it would identify necessary information for sector pruning.

12.6 Sixth priority: sharper height and transfer bounds

The products in Theorem 4.3 are often much smaller than 4^k . Can one characterize the minors that actually attain large values, replace column norm four by a graph-dependent product, or exploit a large totally unimodular submatrix to isolate only a few exceptional quadrilateral columns? The natural parameter may be the number of columns whose addition genuinely destroys unimodularity, not the full occupied support.

Any resulting bound should be checked against the exponential-coordinate layered family. A polynomial coordinate bound in a parameter that remains constant on that family would be false unless the output uses additional compressed representation.

12.7 Seventh priority: reusable source-bound geometry

The supplied-sector API can cache immutable incidence classes, face equations, and boundary homology data for repeated queries on the same triangulation. The independent checker needs either its own cache or an explicit certificate for the shared object. Cache identity must include the exact source and algorithm version. Benchmark full queries with fresh and reused preparation, including A/A controls and certificate replay, before choosing a default.

12.8 Eighth priority: hierarchy interfaces

Extend the algebraic sector description to the constrained surfaces used in hierarchies with boundary patterns. Which additional local normal pieces and weights preserve the small-column-incidence argument? How large can the allowed support, cycle rank, boundary pattern, and proof transcript become after compressed cutting? A local polynomial operation is useful only if the number and size of interfaces across the hierarchy can also be bounded. Lackenby’s current results provide geometric targets for this work [10].

12.9 Ninth priority: careful width-based algorithms

It is tempting to assume that small treewidth of a dual graph solves the remaining search. That implication has not been proved here. The exact positive-Euler normal-surface decision problem parameterized by dual-graph treewidth was explicitly raised as open in Burton’s 2025 Dagstuhl discussion [13]. A width-based state must preserve unbounded integer coordinates, compatibility, canonical vertex-link choices, and the information needed for essentiality. Bounding just the number of graph boundary vertices is insufficient.

12.10 Tenth priority: integration benchmarks that measure the objective

Once diagram provenance and a sector-generation policy exist, run paired whole-recognition tests against the current pipeline. Include easy knots, hard unknot diagrams, nontrivial knots with weak polynomial obstructions, unfavorable sectors, and resource exhaustion. Report failures and overhead, not only successful disc searches. Until then, fixed-sector query improvements should continue to be labelled as such.

13 Research status

The package proves an exact support kernel and a support-dependent primitive height bound, implements candidate discovery and independent replay, and validates the resulting ray sets against a separate topology library. It also proves a representation-specific obstacle to a uniform sparsity strategy. The next decisive advance toward the requested quasi-polynomial recognizer is control of the combinatorial choices of sectors and geometric transitions, with source-bound provenance. The present contribution provides a concrete, tested component for that advance while keeping those remaining obligations visible.

A Exact source and data provenance

The archive’s `results/source_manifest.json` records the GitHub object identities of 175 retrieved files used for the audit and baseline review. The runtime snapshot contains only the unchanged Python runtime files needed for the reproducible package; referenced third-party paper sources are not redistributed. The two new module hashes appear in every final audit result. `MANIFEST.sha256` checks every delivered file except itself.

The experiment outputs are ordinary JSON with decimal integers. The tested coordinates are far below Python’s default large-decimal-string threshold. The native upstream certificate integer codec continues to support its established large-integer encodings. A future public JSON import endpoint should impose input-size and verification budgets explicitly; this research package is a local callable API and reproducible command-line workflow.

The separately retained first benchmark run is exploratory. Only the final source-pinned audit files and the controlled final benchmark provide the counts and measurements quoted in the main text.

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